

PHASE 1 – BRAINSTORMING & IDEATION

1. Project Title

PocketSmart AI: Your Smart Budget & Recommendation Assistant

2. Introduction

Managing personal money can be difficult for students and young people. Many users do not maintain proper records of their daily expenses and may not know where their money is being spent. PocketSmart AI is designed to provide a simple and intelligent solution for managing personal budgets.

3. Problem Statement

Users often struggle to:

Track daily expenses.

Maintain a monthly budget.

Identify unnecessary spending.

Understand their spending patterns.

Plan their savings.

Make informed spending decisions.

4. Proposed Solution

PocketSmart AI is an AI-powered budget assistant that allows users to record their income and expenses. The system analyzes the entered financial information and provides personalized recommendations for budgeting, saving and controlling unnecessary expenses.

5. Target Users

College students

Young professionals

Individuals managing personal expenses

People who want to develop better budgeting habits

6. Key Features

Income tracking

Expense tracking

Expense categorization

Budget management

Spending analysis

AI-powered recommendations

Savings suggestions

Financial summary

7. Expected Benefits

Better expense management

Increased awareness of spending

Easier budget planning

Personalized suggestions

Improved saving habits

PHASE 2 – EMPATHY & USER UNDERSTANDING

1. User Persona

Name: Student User

Age Group: 18–25 years

Occupation: College Student

2. User Needs

The user needs:

A simple way to record expenses.

A clear view of monthly spending.

Help in setting a budget.

Suggestions to reduce unnecessary expenses.

Guidance for saving money.

3. User Pain Points

Forgetting to record expenses.

Spending more than planned.

Difficulty identifying unnecessary expenses.

No clear understanding of monthly spending.

Lack of personalized financial guidance.

4. User Expectations

Users expect the application to be:

Simple

Easy to understand

Fast

User-friendly

Personalized

Accessible from a mobile or computer

5. User Journey

Open Application → Enter Income → Add Expenses → Categorize Expenses → Set Budget → Analyze Spending → Receive AI Recommendations → Improve Spending Habits

6. User Requirements

Functional Requirements

User should be able to enter income.

User should be able to add expenses.

User should be able to categorize expenses.

User should be able to set a budget.

System should calculate spending.

System should provide recommendations.

Non-functional Requirements

Easy-to-use interface

Fast response

Reliable calculations

Secure handling of user data

Clear presentation of information

PHASE 3 – DEFINE THE PROBLEM

1. Problem Definition

Many students and individuals lack a simple tool that combines expense tracking, budgeting and personalized recommendations in one place.

2. Problem Statement

“How might we help students and individuals manage their daily expenses, maintain a realistic budget and make better spending decisions using an AI-powered assistant?”

3. Objectives

The main objectives are:

To develop a simple personal budgeting assistant.

To allow users to record income and expenses.

To categorize expenses.

To monitor the user's budget.

To analyze spending patterns.

To provide AI-powered recommendations.

To encourage better saving habits.

4. Scope of the Project

The project focuses on personal financial management. It helps users understand their spending and provides recommendations based on the information they provide.

5. Challenges Identified

Users may forget to enter transactions.

Users may enter incorrect information.

Spending patterns may change frequently.

AI recommendations depend on the quality of user-provided data.

PHASE 4 – SOLUTION DESIGN

1. Proposed System Architecture

User



PocketSmart AI Interface



Income & Expense Input



Expense Categorization



Budget & Spending Analysis



AI Recommendation Engine



Personalized Recommendations



User

2. Main Modules

Module 1 – User Input

Allows users to enter:

Income

Expense amount

Expense category

Date

Description

Module 2 – Expense Tracker

Stores and displays the user's expenses.

Module 3 – Budget Manager

Allows users to create a monthly budget and compare planned spending with actual spending.

Module 4 – Spending Analyzer

Analyzes spending according to different categories.

Module 5 – AI Recommendation Engine

Uses AI to generate personalized suggestions based on the user's financial information.

Module 6 – Dashboard

Displays:

Total income

Total expenses

Remaining budget

Major expense categories

Savings information

AI recommendations

3. Sample Expense Categories

Food

Transportation

Education

Shopping

Entertainment

Bills

Health

Other

4. Sample AI Recommendations

Reduce spending in high-expense categories.

Set a realistic monthly budget.

Allocate part of the remaining money toward savings.

Review recurring expenses.

Monitor spending regularly.

PHASE 5 – PROTOTYPE DEVELOPMENT

1. Prototype Objective

The prototype demonstrates how the PocketSmart AI application will work before complete development.

2. Proposed Screens

Screen 1 – Home/Dashboard

Displays:

Welcome message

Total income

Total expenses

Remaining balance

Current budget

AI recommendation

Screen 2 – Add Income

Fields:

Income amount

Date

Source

Screen 3 – Add Expense

Fields:

Expense amount

Category

Date

Description

Screen 4 – Budget

User can:

Set monthly budget.

View used amount.

View remaining amount.

Screen 5 – Spending Analysis

Displays spending according to categories.

Screen 6 – AI Recommendations

Displays suggestions generated based on the user's spending information.

3. Prototype Flow

Login/Home → Dashboard → Add Income → Add Expense → Set Budget → Analyze → AI Recommendation

4. Prototype Benefits

Helps identify design problems.

Allows early user feedback.

Demonstrates the application's workflow.

Makes final development easier.

PHASE 6 – DEVELOPMENT & IMPLEMENTATION

1. Technologies Used

Frontend

HTML

CSS

JavaScript

AI

Google Gemini Models / Gemini API

Backend

Python or suitable backend technology

Database

Suitable database for storing user transactions.

2. Development Process

Step 1 – Interface Development

Create the dashboard and required pages.

Step 2 – Expense Management

Implement income and expense entry.

Step 3 – Categorization

Allow expenses to be classified into different categories.

Step 4 – Budget Calculation

Calculate: $\text{Remaining Budget} = \text{Total Budget} - \text{Total Expenses}$

Step 5 – Spending Analysis

Calculate expenses by category and identify major spending areas.

Step 6 – AI Integration

Connect the application with the Gemini model to generate recommendations.

Step 7 – Recommendation Generation

Provide suggestions based on:

Income

Expenses

Budget

Spending categories

Remaining amount

Step 8 – Integration

Connect all modules and test the complete application.

3. Example User Flow

Income: ₹15,000

Monthly Budget: ₹12,000

Expenses: ₹8,500

Remaining Budget: ₹3,500

The AI can use this information to provide budgeting and saving suggestions.

4. Expected Output

The system should display a clear financial summary and personalized recommendations.

PHASE 7 – TESTING & VALIDATION

1. Testing Objective

Testing ensures that PocketSmart AI works correctly and provides reliable results.

2. Test Cases

Test Case

Input

Expected Result

Add income

₹15,000

Income recorded

Add expense

₹500

Expense recorded

Select category

Food

Expense categorized

Set budget

₹12,000

Budget saved

Calculate balance

Budget – Expenses

Correct balance displayed

Generate recommendation

Spending data

AI recommendation displayed

Invalid amount

Text instead of number

Error message

Empty field

No amount

Validation message

3. Functional Testing

Check whether:

Buttons work correctly.

Forms accept valid inputs.

Expenses are stored.

Calculations are correct.

Recommendations are generated.

4. Usability Testing

Check whether users can:

Understand the dashboard.

Add transactions easily.

Find their spending information.

Understand AI recommendations.

5. Validation

The application should be tested using different income, expense and budget values to verify that calculations and recommendations work correctly.

6. Expected Testing Result

The application should perform the required functions correctly, provide understandable information and handle invalid inputs appropriately.

PHASE 8 – DEMONSTRATION & FINAL SUBMISSION

1. Project Introduction

PocketSmart AI is an AI-powered personal budgeting assistant designed to help users track expenses, manage budgets, analyze spending and receive personalized recommendations.

2. Problem

Users often have difficulty tracking their expenses and maintaining a proper budget.

3. Solution

PocketSmart AI combines expense tracking, budgeting, spending analysis and AI recommendations into one application.

4. Demonstration Flow

During the demonstration:

Open the application.

Show the dashboard.

Enter sample income.

Add sample expenses.

Select expense categories.

Set a monthly budget.

Display spending analysis.

Generate an AI recommendation.

Show the final financial summary.

5. Key Features Demonstrated

Income management

Expense management

Budget management

Expense categorization

Spending analysis

AI recommendations

Savings suggestions

6. Project Outcome

PocketSmart AI provides a simple way for users to understand their spending and manage their personal budget with the support of AI-generated recommendations.

7. Future Enhancements

Mobile application

Voice-based assistant

Advanced spending prediction

Personalized savings goals

More detailed financial reports

Notification/reminder system

Additional AI-based financial insights

8. Final Deliverables

Project Documentation

Source Code

GitHub Repository

Project Screenshots

Demo Video

Presentation

Final Project Demonstration

9. Conclusion

PocketSmart AI aims to make personal budgeting easier and more accessible. By combining expense tracking, budget management, spending analysis and AI-powered recommendations, the system can help users understand their financial habits and make more informed day-to-day spending decisions.